PH.40121 **FALL 2026**

General Theory of Relativity II: Extended Adapted Coordinates and the Standard Table

Problem Set 8

Munhyeok Kang Due 2026-10-25 23:59

Topic

Parameter Identifiability in the 1916 Model

Submission

Submit one PDF. Show the derivation, identify every approximation, and include a short consistency check. The maximum file size is 25 MB.

Problems

1. Derive the first displayed relation used in Parameter Identifiability in the 1916 Model. State all assumptions and conventions.

$$rank(F_{ij}) = rank(\sum_a \sigma_a^{-2} \partial_i T_a \partial_j T_a)$$

2. Evaluate a nontrivial case using the second relation. Carry units and uncertainty where applicable.

$$N_\theta = 11, N_{bundle} = 8$$

3. Compare the third relation in two limiting regimes and explain the physical meaning of the result.

$$N_\theta^{2026} = 47, N_O^{2026} = 52$$

Show enough intermediate work that another student can reproduce the calculation.